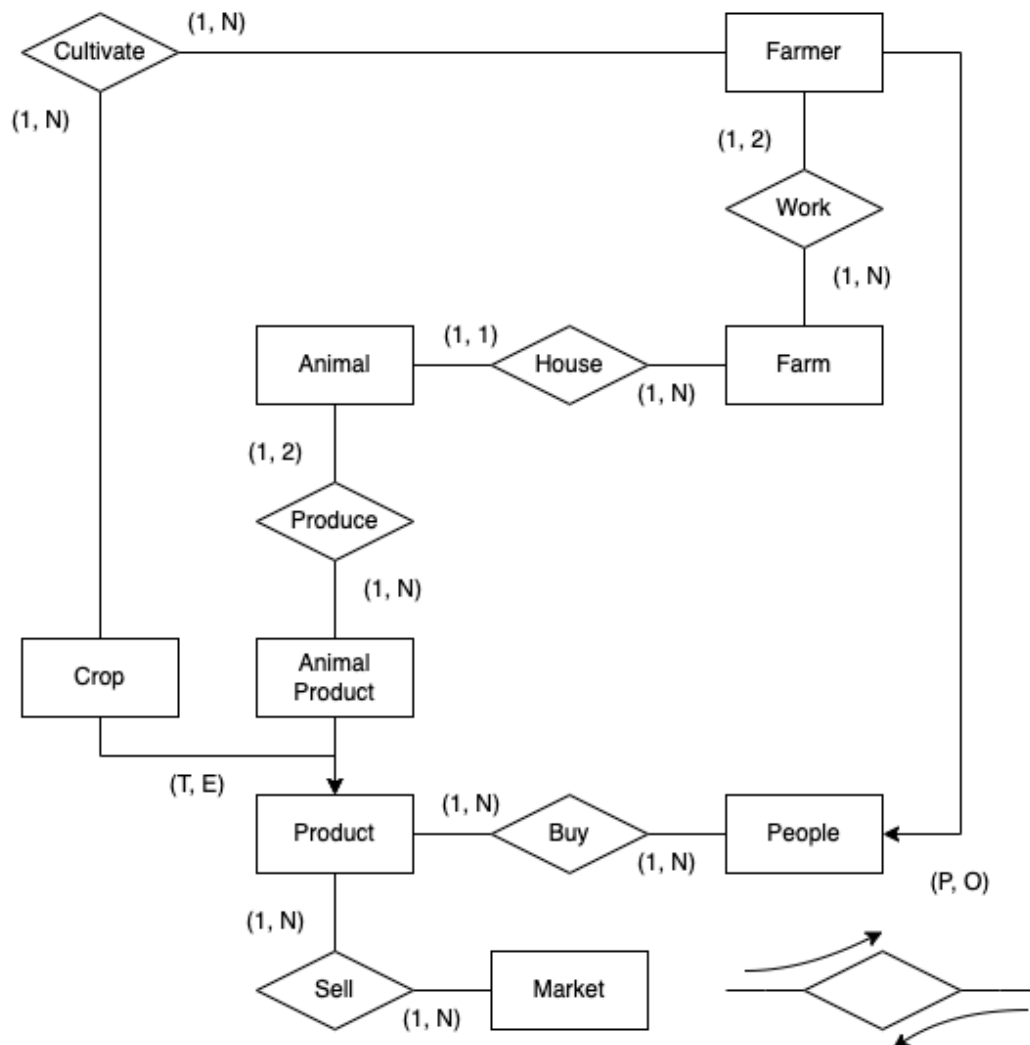


Last Name (Surname, Cognome) _____ First Name _____ Codice Persona _____

Consider the following ER Diagram, representing a data structure for tracking farmers and their sales at local markets.



The following attributes describe the entities. The primary keys are underlined.

- **People** - Personal ID, Name, Surname, Age, Address, State
 - **Farmer** - (...), Biography, Experienced (Yes/No)
- **Farm** - Name, Address, State, Description
- **Animal** - Tag, Name, Age, Animal Type (e.g., Cow), Breed
- **Product** - Product ID, Name, Quantity (in grams), Expiration Date
 - **Animal Product** - (...)
 - **Crop** - (...), Planting Season (e.g., summer), Growth Time (in months)
- **Market** - Name, Location, Opening Time, Closing Time, Size (in m²)

The following attributes describe the relationships.

- **Cultivate** - Planting Date
- **House** - Housing Date
- **Buy** - Buy Date, Amount

Exercise 1 - Cassandra (1 PT)

Suppose you expect to record large-scale, continuously increasing number of sales of products in the market, and therefore you decide to store it in a columnar database. Write the specification of the tables (column families) of a Cassandra deployment assuming you may want to query the data by market location or product name.

Exercise 2 - Neo4j (5 PT)

Consider the entities Product, Crop, Animal Product, Animal, Market, and People (do **not** consider Farmer) from the ER model. Suppose you want to store the respective data instances in a graph database.

Sketch a graph model/example describing the nodes, main attributes, and edges. Either show an example graph or a graph with types. (1 PT)

Write a Cypher query to compute the total number of crops sold for each planting season and each market, considering only markets whose opening time is earlier than 7:00 AM and purchased by people older than 55 years old. (2 PT)

Write a Cypher query to compute, for each person who bought at least one product produced by an animal at least 20 years younger than him/her, the number of different crops they bought after March 20, 2025. (2 PT)

Exercise 3 - Elasticsearch (3 PT)

Consider the Farmer entity. Provide the complete mapping of the index for the considered entity (i.e., field name, field type, the structure of the mapping, etc.) (1 PT)

PUT /

Write a query to return the farmers whose name is "Bill", whose biography contains the words "cactus" and "aloe", and does not contain the word "potato". Additionally, if the "crop" word is found, assign a higher score to the farmer. (2 PT)

Exercise 4 - Redis

Consider the Animal entity.

Create a list "Farm" with two animals named "Bessy" and "Bob" (1 PT)

Write the code to create a new element "Dog" with value "Jerry" that will expire in 1000 seconds. (1 PT)

Write the code to check whether the animal "Bob" is in a list of animals named "Farm" (1 PT)

Exercise 5 - MongoDB (4.5 PT)

Consider the entities Farmer, Farm, Animal, and Crop. How many collections would you define within a documental database (clearly state it)?

How would you implement the relations between the concepts? Provide a simple documental representation, complete with all variables and their types (or an example of a value). (1 PT) **N.B.** Do not add variables besides the ones included in the ER model.

Write a query to compute the total number of crops cultivated by each different farmer whose name is "Rick", returning only the ones who cultivated at least 20 crops. (1.5 PT)

Write a query to compute the total number of farms that house at least 10 cows of at least 5 different breeds. (2 PT)